

priors, which is not straightforward to implement for the Bayes factor package, due to its use of improper prior distributions.

Introduction to marginal simulation-based calibration (SBC). We have recently used marginal SBC for Bayes factors (Schad et al., 2023), which is a statistical technique designed to test whether a Bayes factor estimated in a given analysis is accurate, or whether it is biased and deviates from the true Bayes factor. Here, we first provide a short description of marginal SBC (derived from Schad et al., 2023).

We can formulate marginal SBC for Bayes factors (i.e., for model inference), where $\mathcal{M}$ is a true model used to simulate artificial data y , and $\mathcal{M}'$ is a model inferred from the simulated data. Marginal SBC for model inference makes use of model priors $p(\mathcal{M})$, which are the prior probabilities for each model before observing any data. Marginal SBC can then be formulated as follows (Schad et al., 2023):

$$p(\mathcal{M}') = \sum_{\mathcal{M}} \int p(\mathcal{M}' | y) p(y | \mathcal{M}) p(\mathcal{M}) dy . \quad (6)$$

We can read this equation sequentially (from right to left): first, we sample a model from the model prior, $p(\mathcal{M})$. Next, we simulate data based on this model, $p(y | \mathcal{M})$. This in fact involves two steps: simulating parameters from the parameter prior, $p(\Theta | \mathcal{M})$, and then simulating artificial data from the parameters and model, $p(y | \Theta, \mathcal{M})$, i.e., $p(y | \mathcal{M}) = \int p(y | \Theta, \mathcal{M}) \times p(\Theta | \mathcal{M}) d\Theta$. Next, we estimate the posterior model probabilities from the simulated data, $p(\mathcal{M}' | y)$. This again involves several steps: we compute marginal likelihoods, $p(y | \mathcal{M}')$, for both models, compute Bayes factors, and compute the posterior probability for each model given the data, $p(\mathcal{M}' | y)$, by adding the model prior. That is, we obtain a posterior model probability for each simulated data set. We can now compare, whether the posterior model probability, $p(\mathcal{M}' | y)$, – averaged across all simulated data sets – is the same as the prior model probability, $p(\mathcal{M})$.

The key idea is that if the computation of Bayes factors and posterior model probabilities is performed correctly (and of course the data simulation is implemented correctly), then the average posterior probability for a model should be the same as its prior probability. By contrast, if the average posterior probability for a model deviates from its prior probability, then this indicates that the Bayes factor estimate is biased, i.e., that the obtained Bayes factor does not correspond to the true Bayes factor.

For a given run of marginal SBC, to test whether the Bayes factor estimates are biased, we perform Bayesian t-tests on the posterior model probabilities, i.e., computing null hypothesis Bayes factors to test whether the posterior model probabilities differ from the prior model probabilities. Specifically, we perform paired Bayesian t-tests, where for each simulation, we compare the posterior model probability (taking values between 0 and 1) against the prior sampled model (taking values of either 0 or 1). We test the two hypotheses that the difference between prior and posterior is zero (H_0), or is not zero (H_1) and that the standardized effect size of the difference follows a Cauchy distribution with default scale parameter of $\sqrt{2}/2$. We moreover perform sensitivity analyses, where we vary the scale parameter to study its influence on the results. We perform these t-tests using the R-package `BayesFactor` (function `ttestBF`) to test whether SBC indicates bias in the estimation of Bayes factors.

We note that the marginal SBC described here (i.e. that data-averaged posterior equals the prior) is a different check than what is commonly referred to as SBC in the literature, (sometimes called *conditional* SBC to make the difference explicit). See Modrák et al. (2023) for a comparison between the marginal and conditional formulation as well as review of other related approaches to Bayesian model checking. Conditional SBC is sensitive to different computation failures than marginal SBC. A related desirable property of Bayes factor is that the posterior model probabilities are calibrated, i.e. that whenever $p(\mathcal{M}_i | y) = q$, the true model should be $\mathcal{M}_i$ in q of the cases. This calibration can be directly tested with a range of

approaches, here we will use diagrams described in (Dimitriadis, Gneiting, & Jordan, 2021). In fact, this form of calibration is equivalent to conditional SBC in a suitable reformulation of the problem. Proof of the equivalence and more discussion will be presented in an upcoming paper (Modrák & Bürkner, in preparation).

Methods

We performed marginal SBC analyses for a range of experimental designs. Here, we report the methodological details of these simulation analyses.

Design 1: 2x2 design with random effects for subjects

We first performed marginal SBC for Bayes factors in a 2×2 design inspired from the 2-step decision-making task (Daw, Gershman, Seymour, Dayan, & Dolan, 2011). In this task, the probability to repeat an action is examined as a function of whether the last trial was rewarded and whether during the last trial a common or a rare transition occurred in the task. The task is designed to investigate the influence of model-free versus model-based reinforcement learning strategies on human learning and decision-making, where a model-free strategy is captured by the main effect of reward, whereas a model-based strategy is captured by the interaction reward $\times$ transition probability. Here, we assume that the decisions to repeat an action are averaged for each of several blocks of trials, yielding an approximation to a normal distribution.

In the 2×2 repeated measures design, each of the 4 conditions was repeated 5 times/blocks, leading to 20 (averaged) data points per subject. We simulated data from 10 subjects, yielding a total of $20 \times 10 = 200$ data points. We simulated data from the following model (H1), where n indicates the row of the data frame, and $subj[n]$ yields the subject number in row n .